WASP-50b

R-1521

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_150114` · created 2026-08-01T17:12:46+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457036.624 +/- 0.056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.048
Transit depth (Rp/Rs)^2	3.07 +/- 1.68 [%]
Orbital Inclination (inc)	89.71 +/- 2.55
Airmass coefficient 1 (a1)	16.38 +/- 4.06
Airmass coefficient 2 (a2)	-1.96 +/- 0.19
Scatter in the residuals of the lightcurve fit is	17.05 %
Best Comparison Star	#2 - [92, 29]
Optimal Aperture	5.59
Optimal Annulus	9.21
Transit Duration (day)	0.05 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.048 vs 0.1404 (deviation 0.26σ)
Duration fit vs published	0.05 d vs 0.0754 d (deviation 0.27σ)
Mid-time O–C	-56.0 min
Depth SNR	1.3
Residual scatter	17.05 %

Light curve

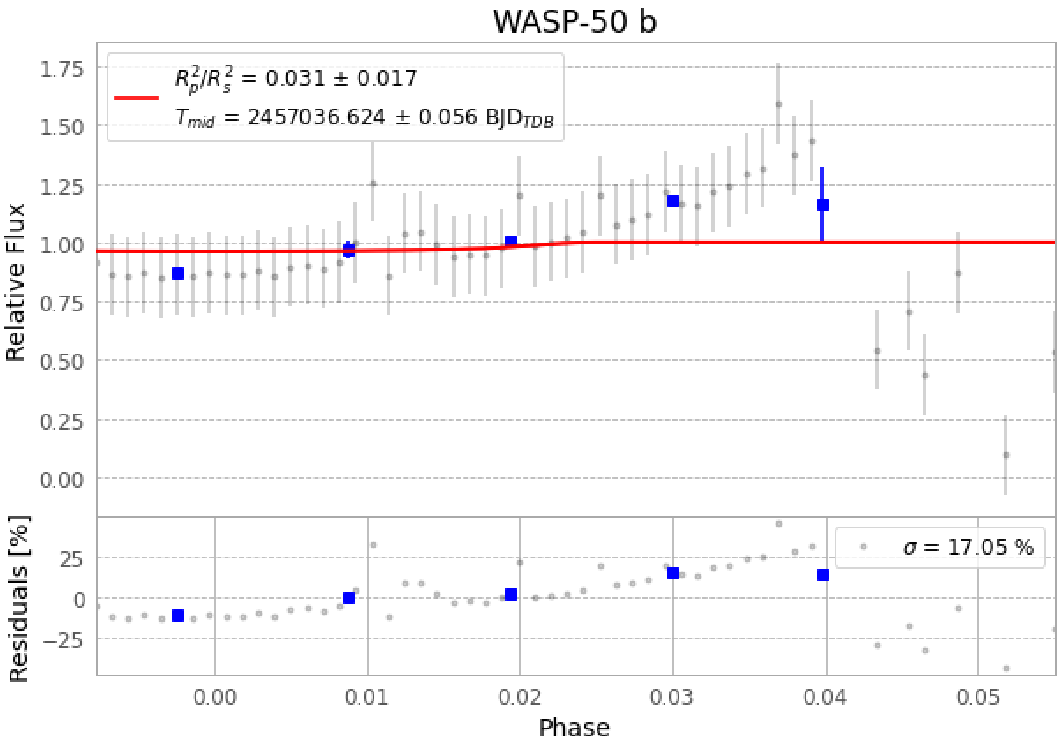

Figure 1 — FinalLightCurve_WASP-50 b_2015-01-13.png (SHA-256 142db3d39e7fb732…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [202, 314], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 be6de098e077697d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

67 FITS frames (44 MB), `WASP-50150114023312.FITS` → `WASP-50150114055112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-150114.log (31dde0a4c229…), FinalLightCurve_WASP-50 b_2015-01-13.png (142db3d39e7f…), FinalLightCurve_WASP-50 b_2015-01-13.csv (0744c6bade52…)

Compute resources

Wall time 220.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)